

Estimation of bankruptcy risk in the Valuatium platform



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 - Random forest
 - Artificial neural network
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Introduction

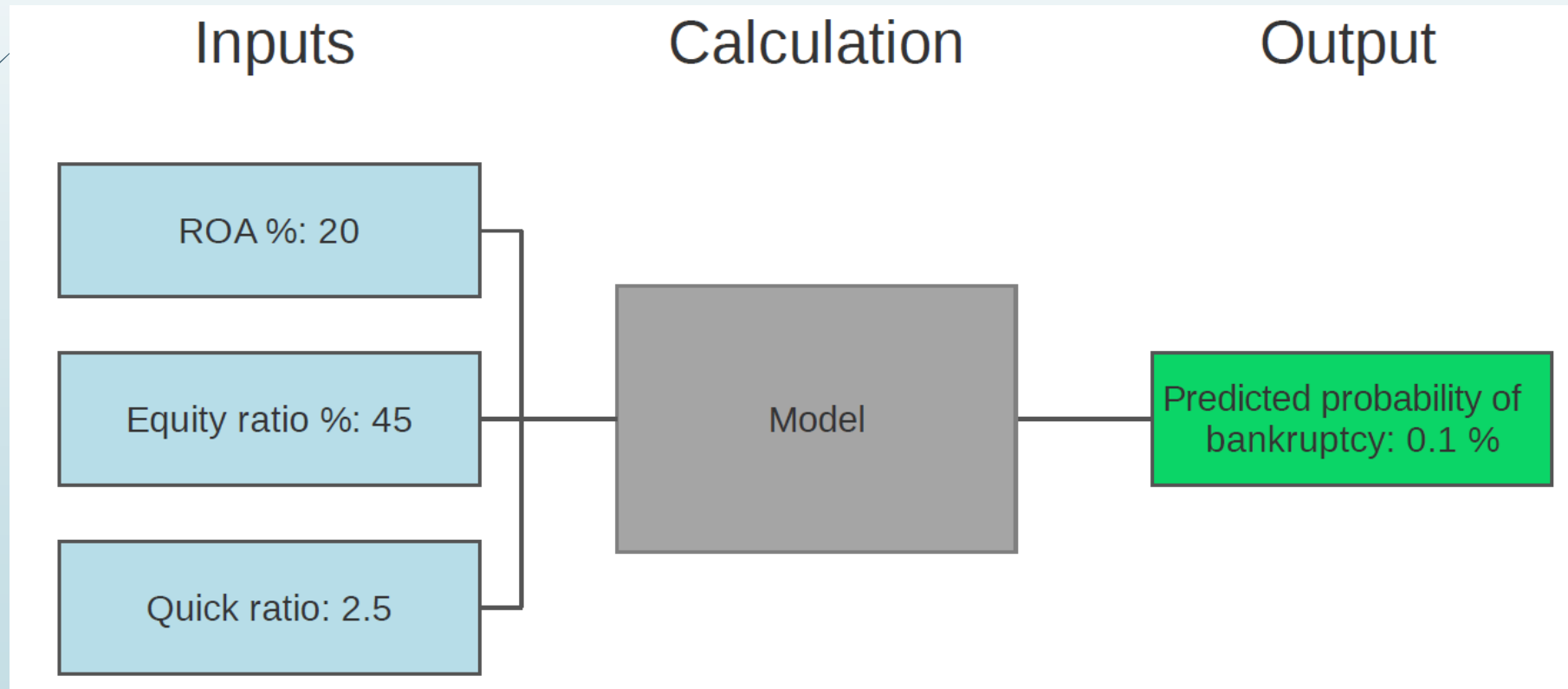
- Valuatum's bankruptcy model is built using financial data of approximately 170 000 Finnish companies
- Comparison of historical bankruptcy data and company financials reveals the most important factors that can be used in predicting bankruptcies
- Both raw figures (e.g. net sales) and derived financial ratios (e.g. equity ratio) can be used
- The model also considers industry-specific risks
 - i.e. certain industries are more prone to bankruptcies

Development of the bankruptcy model 1/2

- The first step of model development is training the model
- The model is trained by inputting a large amount of financial data (input variables) and information about occurred bankruptcies (output)
- Based on input variables, the model estimates the probability of bankruptcy for a given company
- Model performance can be tested using a test set
- Different samples of the same data set (approx. 170 000 Finnish companies) are used for training and testing the model
 - This ensures that the training set and the test set are statistically similar
- The model is then optimized in a way that minimizes the error in bankruptcy prediction
- Valuatium platform combines the use of several machine learning techniques and neural networks to predict bankruptcy risk as accurately as possible

Development of the bankruptcy model 2/2

- The result of the training process is a model that takes financial data as input and gives the probability of bankruptcy over the next 1-2 years as an output.



Overview of the used methods

Logistic regression

- A relatively simple method for bankruptcy risk estimation
- Similar to classic linear regression
- The most commonly used model in bankruptcy prediction
- Not currently under development due to superiority of other models

Random forest

- A model that is built based on decision tree learning, a common method used in machine learning
- An input is fed into a tree-like structure where each branch represents a decision
- Several trees are built and together they form a random forest

Valuatum Platform

Artificial neural network

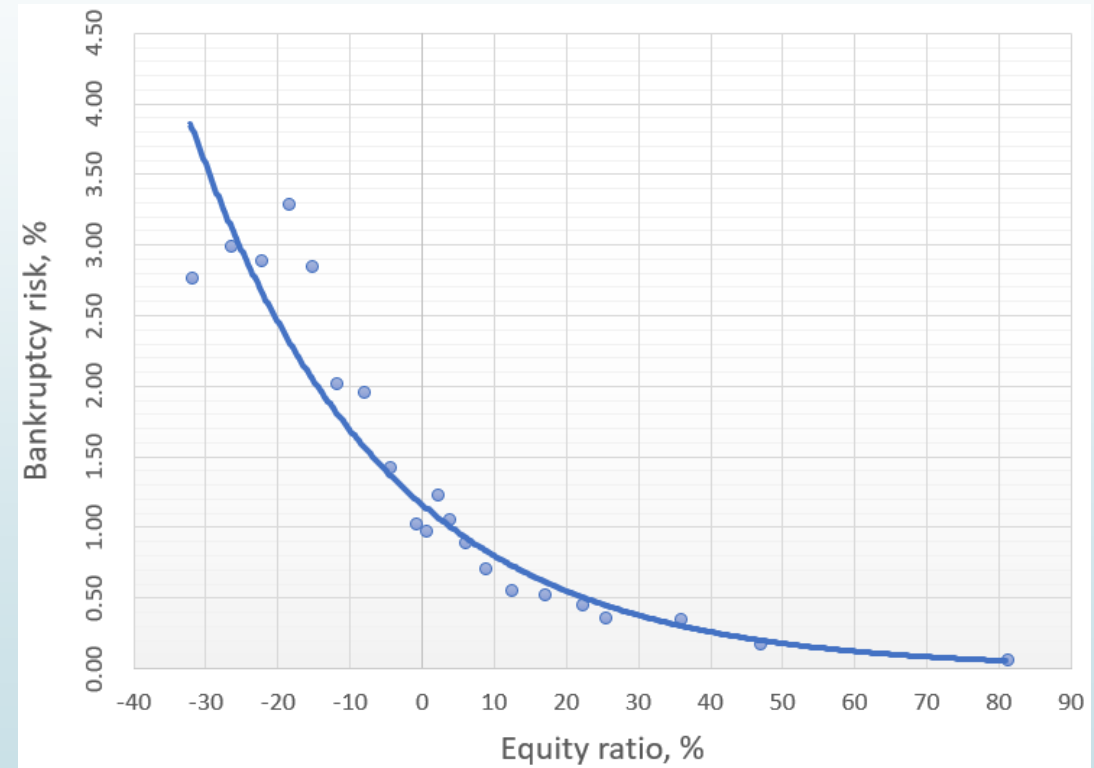
- An input is fed into a network where it runs through several layers. Each layer makes adjustments until an output layer is reached and a prediction is made
- The process is repeated until the prediction is close to the real outcome
- Artificial neural network is analogous to a human brain

Ensemble method

- Combination of all aforementioned methods
- Captures the upsides of all methods but the results are often difficult to interpret
- Reduces the effect of errors made by individual models

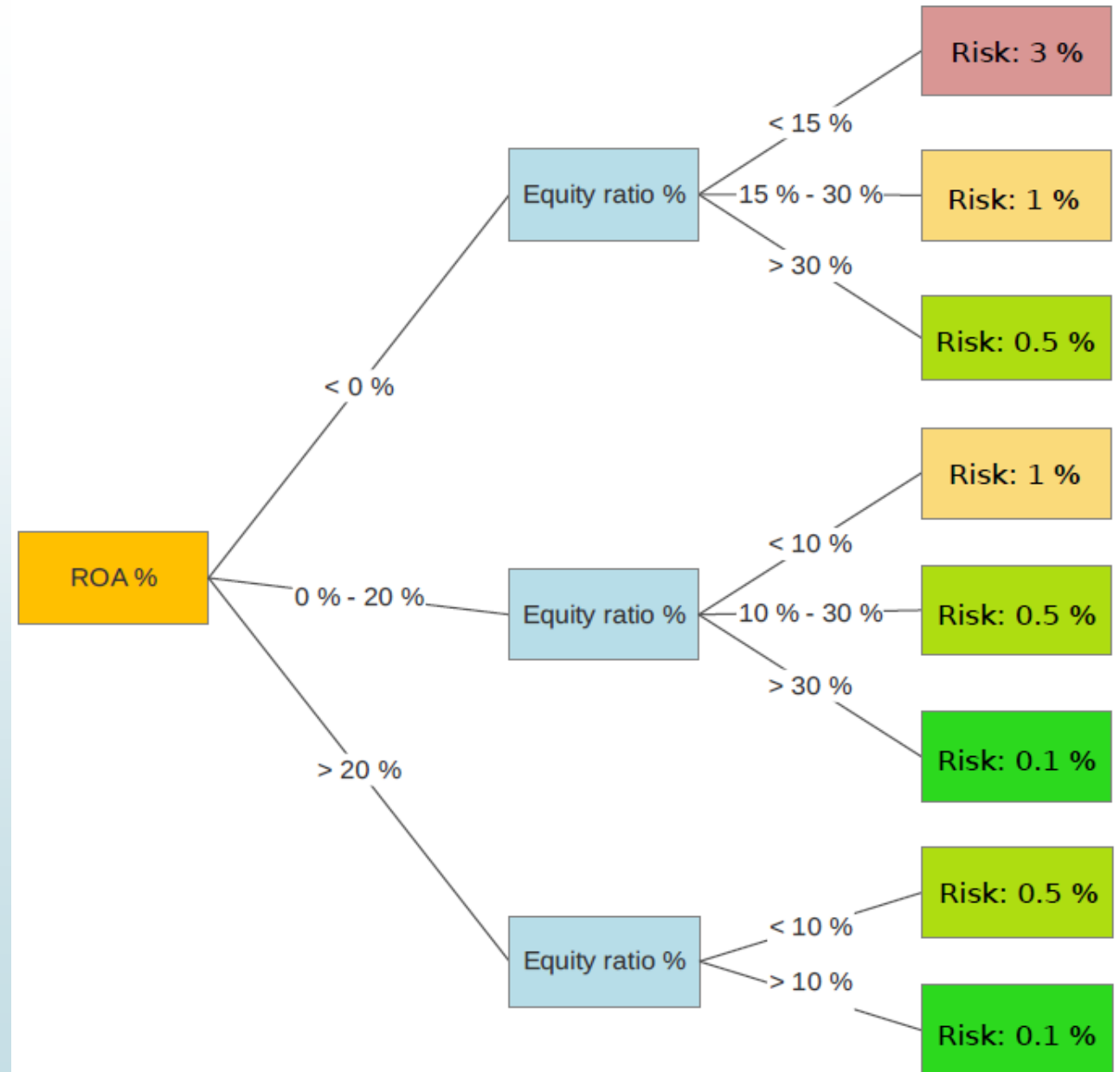
Logistic regression

- Similar to linear regression
 - Each input a_i is assigned a weight w_i and the results are summed together
 - The result is then passed through a logistic function which gives a percentage as an output
- Each input's contribution to the output is easily interpreted
- Theoretically well-suited for bankruptcy estimation but in practice quite limited
 - Logistic regression assumes that each input contributes to the outcome linearly and that the inputs are uncorrelated



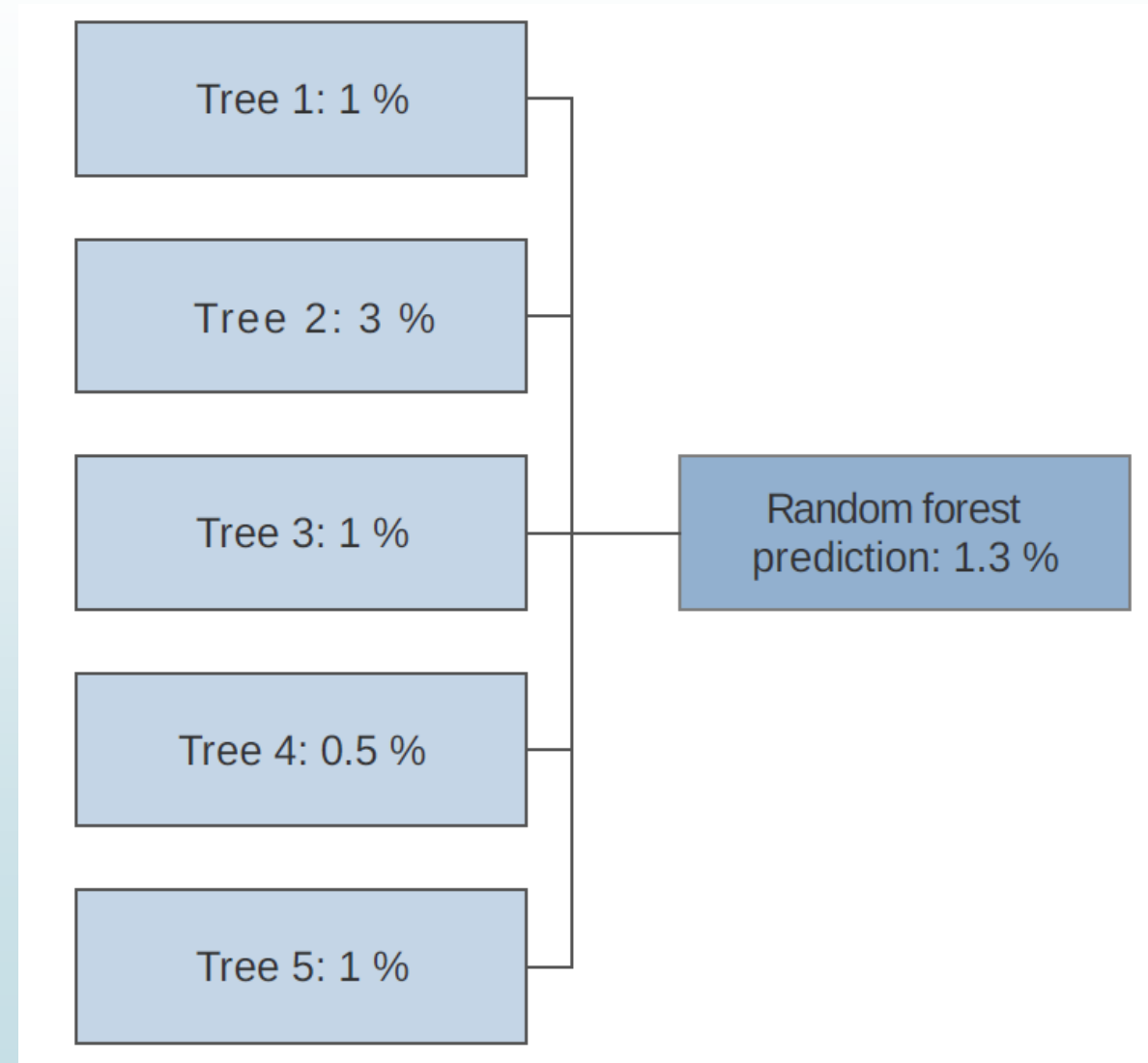
Random forest 1/2

- A random forest model is built using decision trees where each node consists of a test on one of the input attributes (e.g. whether ROA is positive or negative)
 - Each branch represents an outcome of a test
- At the end of each branch is a decision (= probability of bankruptcy)
- When training a decision tree, each node is set so that estimated bankruptcies match the actual bankruptcies as closely as possible



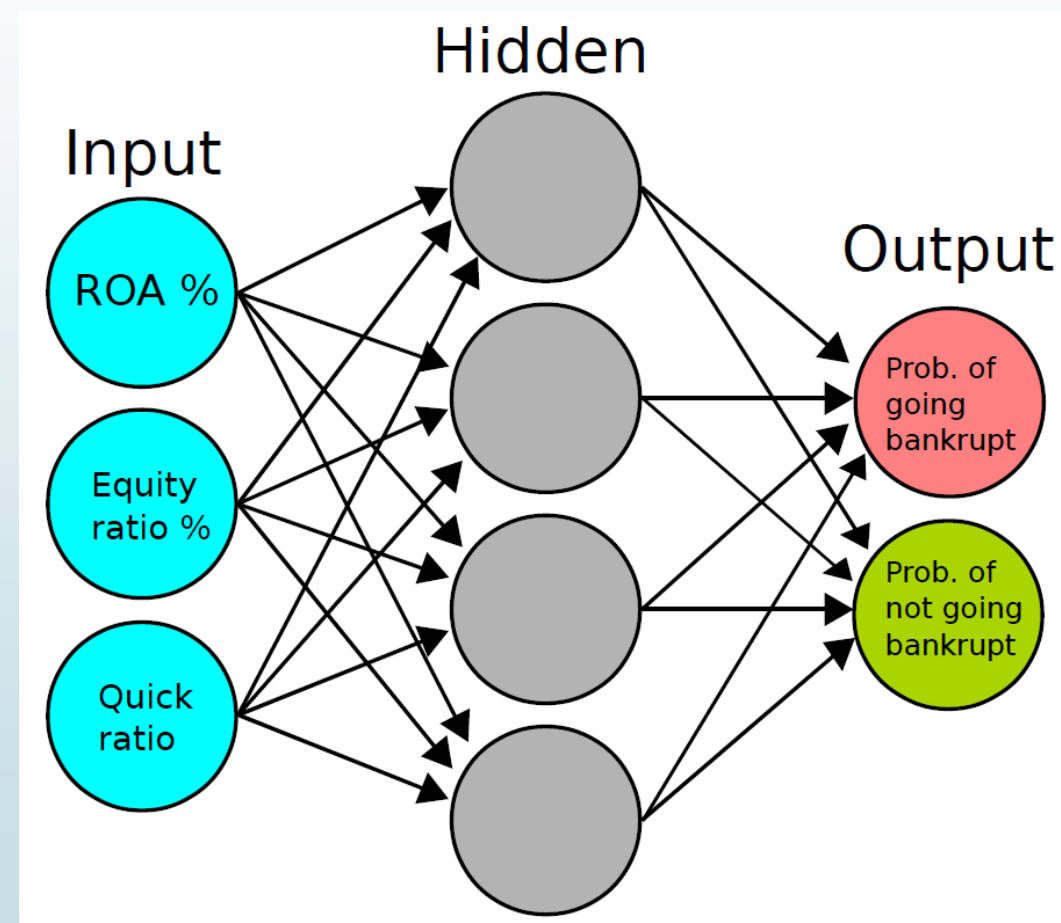
Random forest 2/2

- A random forest consists of hundreds or thousands of decision trees
- Each individual decision tree
 - has a different structure
 - is trained with a different subset of data
 - has "a vote" on how likely a company is to go bankrupt
- The average of all decision trees forms the random forest prediction
- The random forest model can learn very complex relationships, but the results are not easy to interpret (i.e. which inputs affect the output)



Artificial neural network 1/2

- An artificial neural network is based on a collection of nodes called neurons that are connected to each other. Each neuron receives data as input.
- The network consists of several layers where neurons are located
 - The first layer is called an input layer where the original data is fed into the network
 - The last layer is called an output layer where the bankruptcy risk is calculated
 - Between the input and output layers, there are one or more hidden layers
- In each layer, the inputs are weighed and summed, and passed through an activation function, which enables the modeling of non-linear dependencies
- The output of the activation function is then passed on as an input to the next layer of neurons

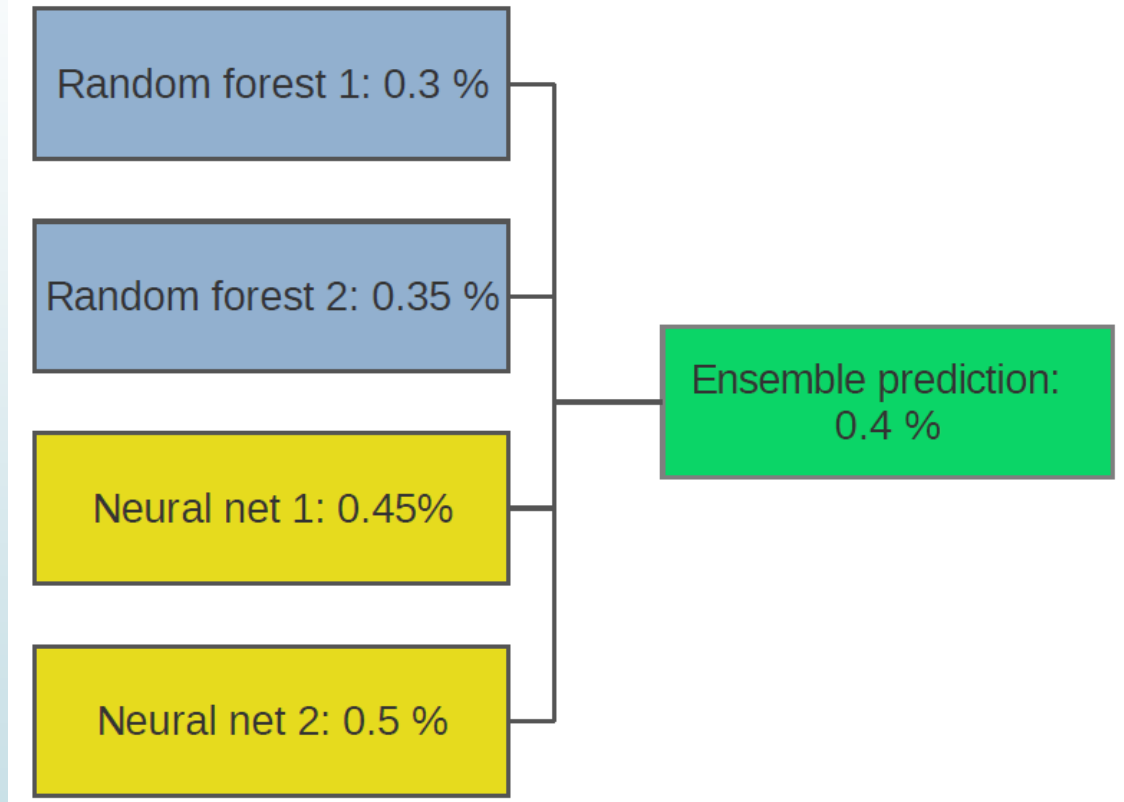


Artificial neural network 2/2

- The neural network is trained using a backpropagation algorithm
 - Each neuron is assigned a weight which is adjusted throughout the training process
 - The goal is to find weights that result in a predicted probability of bankruptcy that is as close as possible to the actual state of the company
 - If the error in the outcome is too large, the output is fed back to the network and the process is repeated
- The output of the neural network can be interpreted as a probability, provided that the activation function is a logistic function
- Neural networks can learn very complex relationships, but the results are difficult to interpret due to the obscurity of the adjustments made in the hidden layers

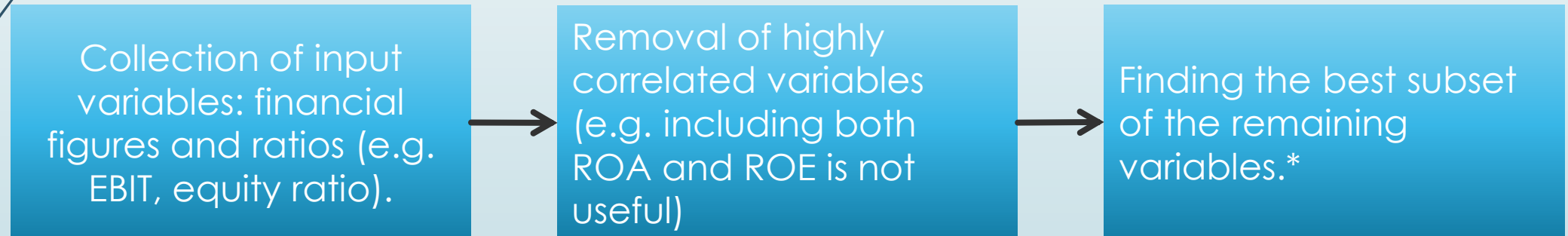
The ensemble method

- Different methods give different results even if they are trained using the same data set
- Different methods also have varying strengths and weaknesses
- The ensemble method can be used to combine all the aforementioned models and their upsides
 - We can, for example, train several random forest models and combine them with several neural networks. The output is the average of all the models.
- The downside is that the ensemble model requires more computing resources. It is also impossible to deduce the individual contributions of inputs to the output value



Selection of input variables

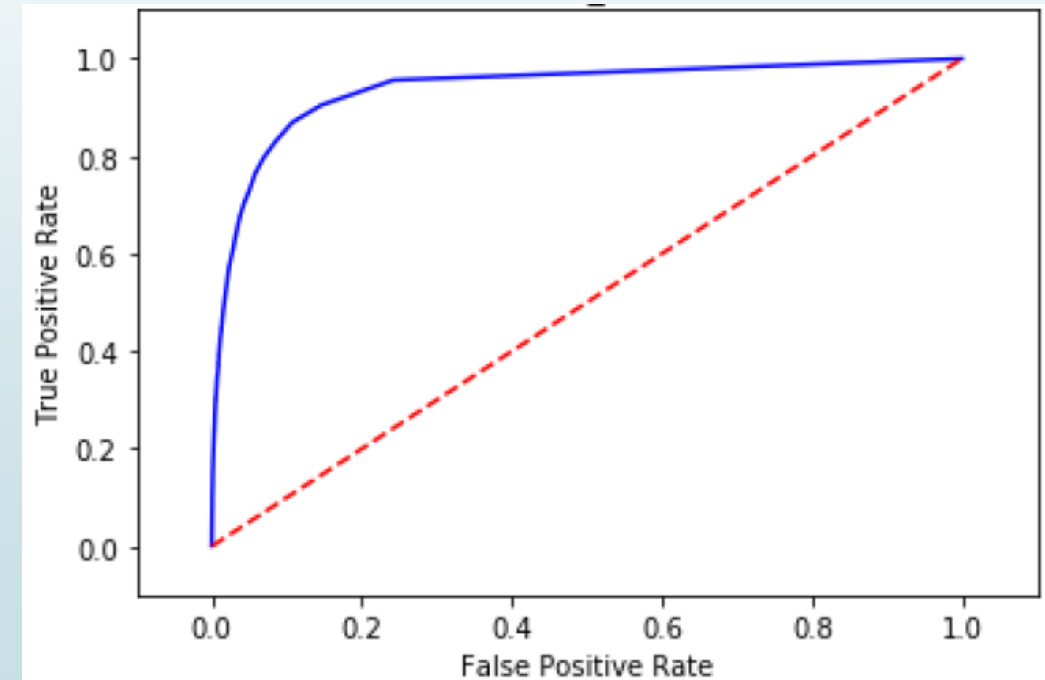
- When creating a statistical model, it is crucial that the model is given relevant data during the training phase
 - It is also important to remove unnecessary variables to reduce variance and speed up computation
- The selection process is described below



** The nature of the last step depends on the model used. In the case of random forest, the model automatically removes unnecessary variables. In the case of logistic regression and neural networks, statistical methods such as stepwise algorithms can be used.*

Performance evaluation 1/2

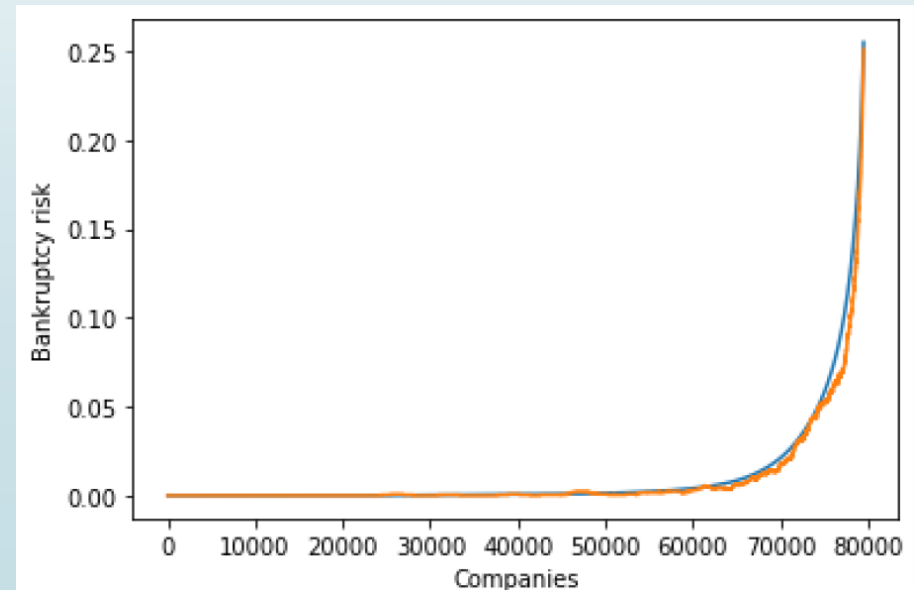
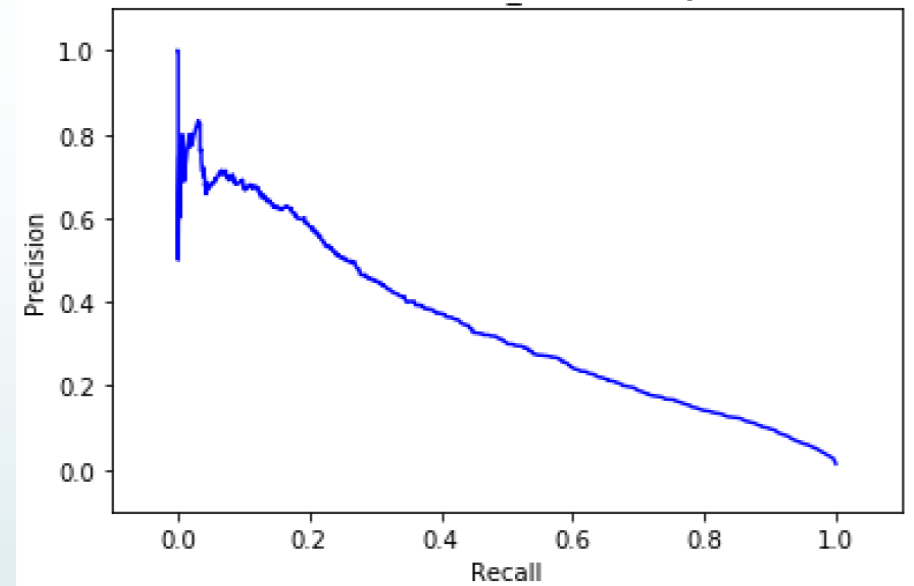
- Accuracy alone does not tell the whole truth: If 1 % of all companies go bankrupt over a given year, a model predicting that no bankruptcies occur would have an accuracy of 99 %. Therefore, other methods are needed to evaluate the performance of the models.
- Receiver operating characteristic (ROC) curve
 - A probability threshold is selected and used to classify companies
 - E.g. if a threshold of 5 % is chosen, then each company with a bankruptcy risk higher than 5% is predicted to go bankrupt
 - True positive and false positive rates are calculated for different thresholds ranging from 0 % to 100 % and plotted to form the ROC curve
 - Area under curve (AUC) can be used to present the result numerically



Performance evaluation 2/2

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- Precision-recall (PR) curve
 - Similar to ROC curve, but instead of true and false positive rates, PR-curve is formed by plotting precision and recall
 - How many of the predicted bankruptcies actually happened (precision) vs. how many of the actual bankruptcies were predicted (recall)
- Performance can be evaluated by calculating AUC
- Moving average graph
 - Data sets are divided into groups of 2500 companies and organized from lowest bankruptcy risk to the highest
 - Moving average of the predicted bankruptcy risk (blue) and actual bankruptcy rate (orange) are plotted
 - Confirms that, on average, the predicted risk corresponds to the actual risk



Results

- A total of approximately 170 000 Finnish companies and 30 input variables were used in the development of the models
- One half of the data was used for training the models, while the other half formed the testing set
- The results are presented below and compared to the results obtained by Altman et al. (2014)*
 - A study based on approximately 2.6 million companies from more than 30 countries using financial data from 2007-2010
- The maximum value for both ROC-AUC and PR-AUC is 1.0. However, the values are not directly comparable
 - ROC-AUC of 0.8 can be considered good, while values exceeding 0.9 are excellent. A random model has a ROC-AUC of 0.5
 - PR-AUC is more difficult to interpret, and there is no general guideline for what can be considered good

	Random forest	Artificial neural network	Ensemble method	Altman et al. (2014)
ROC-AUC	0.904	0.880	0.902	0.894
PR-AUC	0.189	0.146	0.192	-

* Altman et. al. (2014), "Distressed Firm and Bankruptcy prediction in an international context: a review and empirical analysis of Altman's Z-Score Model", Available [online]: <https://pdfs.semanticscholar.org/257c/b4227101b4da636e90b323736c68c0653a4f.pdf>

Contact

- Esa Mäkeläinen, CEO
 - esam@valuatum.com
 - +358 50 307 4880
- Office
 - www.valuatum.com
 - myynti@valuatum.com
 - +358 45 123 0308